

STAT 5720

Open Topics in Statistics and Data Science

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Data Scientist (38.8 / 100)	31.9 / 100

32	64	+48
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Provides a vehicle for the development and presentation of new topics that may be incorporated into the core courses in applied mathematics. Department enforced prerequisite: variable, depending on the topic, see instructor.
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INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	38.8	66.3	+27.5	MEDIUM
Applied Scientist	37.3	85.6	+48.3	HIGH
ML Engineer	33.5	61.7	+28.2	MEDIUM
AI Engineer	32.4	60.5	+28.2	MEDIUM
Machine Learning Engineer	31.6	59.8	+28.2	MEDIUM
MLOps Engineer	18.1	49.6	+31.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS



This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

Data Scientist

38.8 → 66.3 (+27.5 pts)

data sets problems

Applied Scientist

37.3 → 85.6 (+48.3 pts)

computational research reasoning interdisciplinary problems models technical data

ML Engineer

33.5 → 61.7 (+28.2 pts)

data learning machine models model world problems technical

AI Engineer

32.4 → 60.5 (+28.2 pts)

following data models learning software technical model teams

Machine Learning Engineer

31.6 → 59.8 (+28.2 pts)

data sets learning tools design science models model

MLOps Engineer

18.1 → 49.6 (+31.5 pts)

data model learning machine tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.